

Full-Stack Engineer Assignment: Real-Time Chat App

This assignment challenges you to build a basic real-time chat application using the following technologies:

- **Frontend:** Next.js
- **Backend:** Strapi
- **Real-time Communication:** Socket.io

Functionality:

The application should allow users to:

- Register and login with a username.
- Join a chat room with a designated name.
- Send and receive chat messages in real-time within the joined chat room.
- See a list of currently active users in the chat room.

Implementation Details:

- **Frontend (Next.js):**
 - Create separate components for user registration/login, chat room interface, and individual chat messages.
 - Utilize Next.js functionalities like routing and data fetching to manage user authentication and chat data retrieval.
 - Implement Socket.io client-side to establish a connection with the backend and handle real-time communication.
- **Backend (Strapi):**
 - Define a Content-Type in Strapi for storing chat messages.
 - Implement API endpoints for user registration/login and chat message creation/retrieval.

- Utilize Strapi webhooks to trigger Socket.io server-side events when new messages are created.
- **Real-time Communication (Socket.io):**
 - Use Socket.io server-side to establish a real-time connection between the frontend and backend.
 - Implement event listeners on the server to handle new message creation and broadcast messages to all connected clients in the same chat room.

Additional Considerations:

- Implement basic user authentication using username and password.
(Optional: Implement social login with providers like Google or Facebook)
- Style the chat interface using CSS frameworks like Bootstrap or Tailwind CSS.
- Implement basic error handling and user feedback mechanisms.
- Consider storing chat history in a database for future reference. (Optional)

Deliverables:

- A functional Next.js application demonstrating the chat functionality.
- A Strapi project containing the necessary models and API endpoints.
- Documentation outlining the implemented functionalities and code structure.

Evaluation Criteria:

- Code Quality and Readability
- Functionality Completeness
- Implementation of Best Practices (Next.js, Strapi, Socket.io)
- User Interface Design and Usability
- Overall Project Structure and Documentation

Additional Notes:

This assignment is intended to assess your understanding of the mentioned technologies and your ability to build a real-time application. You can feel free to utilize additional libraries or frameworks as needed to enhance the functionality or user experience.

Time Estimate:

This assignment is estimated to take approximately 8-12 hours to complete, depending on your experience level.

Good luck!